

What's New in SentinelWire™ v1.1

(New Capabilities Explained)

A. Time-Based Beaconing Detection (NEW)

SentinelWire™ v1.1 introduces **temporal analysis** to detect automated outbound communication patterns commonly associated with **C2 infrastructure**.

The extractor now evaluates:

- Repeated connections to the same destination
- Average time interval between connections
- Interval regularity (standard deviation)
- Persistence over time

Highly regular, low-variance communication is flagged as **potential beaconing behavior**.

B. Beacon Confidence Scoring (NEW)

A dedicated **Beacon Score (0–100)** is calculated using:

- Connection frequency
- Temporal regularity
- Duration of activity

This score contributes directly to the overall IOC confidence score.

Interpretation

- 0–29 → Low confidence
- 30–69 → Medium confidence
- 70–100 → High confidence (probable automated beaconing)

C. Lateral Movement Detection (NEW)

SentinelWire™ now detects **internal network abuse**, focusing on:

- **SMB (TCP/445)** — peer-to-peer access attempts
- **LLMNR (UDP/5355)** — name resolution abuse
- **NBNS (UDP/137)** — legacy discovery traffic

These signals often indicate:

- Post-compromise activity
- Credential harvesting attempts
- Internal reconnaissance

New IOC types:

- lateral_smb
- lateral_llmnr
- lateral_nbns

D. Enhanced MITRE ATTACK Mapping

v1.1 strengthens behavior-based MITRE alignment, including:

- **T1071.001 / T1071.004** — Application Layer Protocols
- **T1573** — Encrypted Channel
- **T1021.002** — SMB / Remote Services
- **T1557.001** — LLMNR/NBT-NS Poisoning
- **T1046** — Network Service Discovery